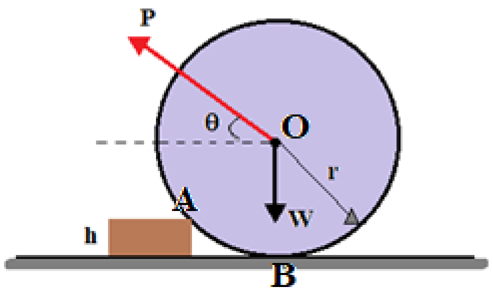
2020-2021 Fall Semester

MTM3701 MECHANICS 1 (Gr.2)

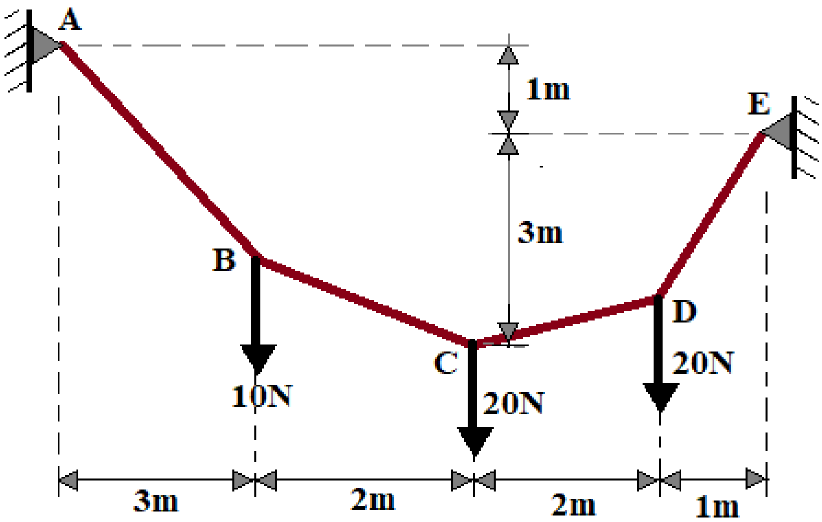
<u>Final Exam</u>

(11.01.2021)

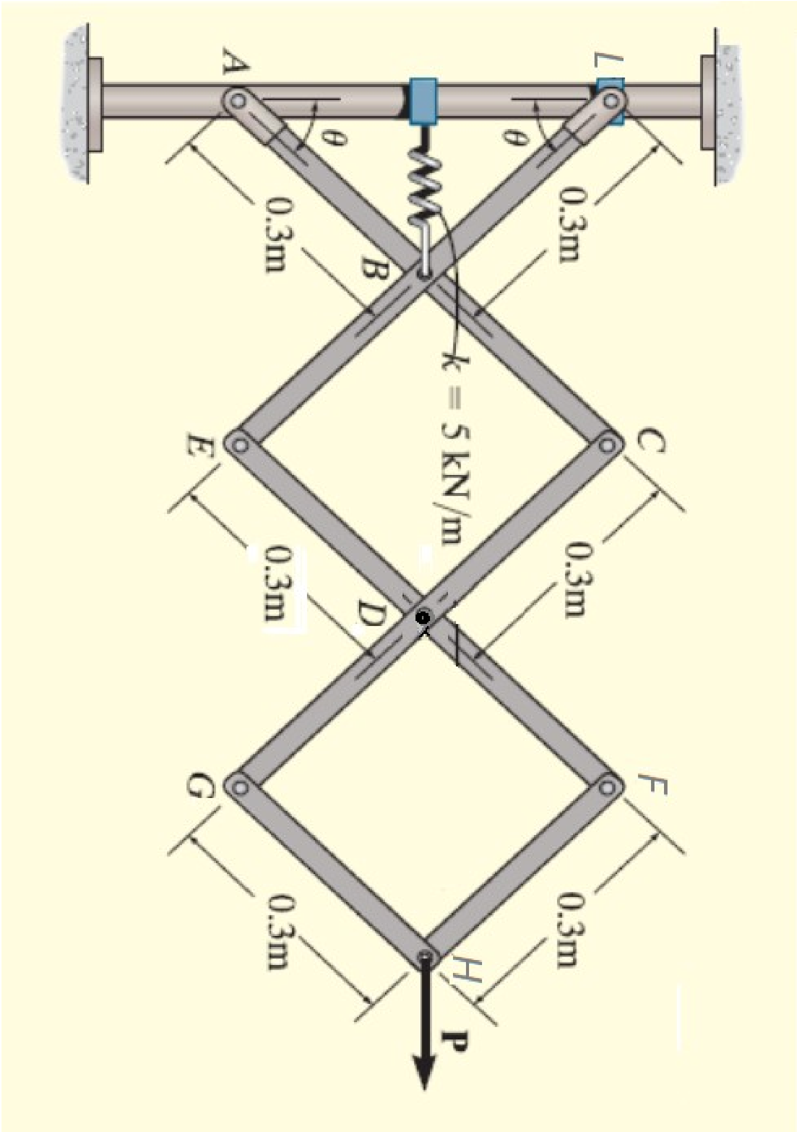


Q-1. (33 Points) Find the smallest value that the force **P**, which is applied at the center **O** of a roller of weight W and makes an angle θ with the horizontal, to overcome the curb of height **h** in front of the roller of radius **r** given in the figure (**h**, **W**, **r** and θ are known) .

Q-2. (33 Points) The cable given in the figure is fixed at points A and E with a fixed support and the individual forces given act on certain points of the cable. According to the given: find the **support reactions**, **the part of the cable with the greatest cable tension** and **its value**, and **the coordinates of the points where the individual loads act**.



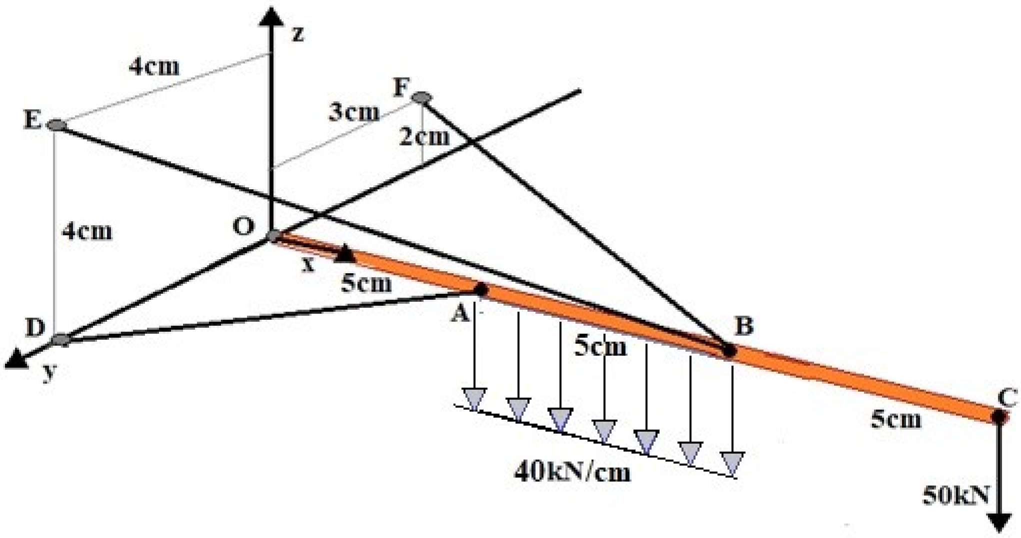
Q-3. (33 Points)



Find the condition that θ should satisfy in the equilibrium state of the mechanism given in the figure by using **Principle of Virtual Work**. The spring is unstretched when $\theta = 0$. Neglect the mass of the links.

<u>**Note**</u>**:** A is a **pin** and L is a **collar on smooth rod**. All contact points are frictionless. A-E-G, B-D-H and L-C-F are on the same direction in the vertical.

Q-4. (33 Points)



The OC rod given in the figure above is in horizontal position, and it is attached to the ground with cables from A and B points and a ball-and-socket joint from O point. There is a ring at point B and the EBF cable is attached to points E and F passing through the ring at point B. A vertical force of 50kN acts on the rod from point C and a uniformly distributed load of 40kN/cm is attached to the piece AB. Find the support reaction at O and the EBF and AD cable tensions for the given loading situation. Neglect the weight of the rod and cables.

Note: **You are required to answer any 3 questions!**

Answers will be resolved cleanly and neatly, Name, Surname and number will be written in the upper left corner of the first page.

Duration: 100 min.

(The last 10 minutes are given for online upload).